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ADYAPAN

Adyapan School

Drug Design

Duration - 2 months

Industry Certification

Skill India Certified

250+ Partner Companies

From chemistry to cure Master modern drug discovery.

From molecules to medicines - become industry-ready.

This immersive Drug Design program takes you beyond foundational concepts into real-world pharmaceutical discovery workflows. Learn molecular modeling, structure-based drug design, virtual screening, and ADMET analysis through hands-on tools and guided case studies. With a strong focus on computational techniques and industry-relevant practices, you'll graduate with the ability to design and evaluate drug candidates, analyze biological targets, and contribute to modern drug discovery pipelines with confidence.

8	30+	20,000+
MODULES	PROGRAM OFFERINGS	STUDENTS

250+ PARTNERED COMPANIES

ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch your career.

Adyapan Schools was built with a single conviction: learning works best when it happens in the real world. We partner with top companies, mentors, and industry platforms to ensure every student graduates with a portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live client projects, giving you the skills and proof-of-work that employers actually want.

MISSION

To equip ambitious learners with practitioner-level digital marketing skills through mentor-led, project-based education that bridges the gap between learning and earning.

VISION

To be India's most trusted launchpad for the next generation of marketing leaders — defined not by degrees but by the real work.

Adyapan School

Everything you need to grow fast

PROGRAM HIGHLIGHTS

Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.

1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.

AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.

Dual Certification

Earn both a Course Completion and Internship Certificate — accredited by Skill India Digital Hub and NSDC.

Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.

Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 1

Introduction to Drug Discovery & Target Identification

- Discussion of curriculum
- Overview of the modern drug discovery pipeline
- Understand the concept of a drug target
- Explore disease biology and how dysregulated pathways reveal druggable targets
- Introduction to target classes
- Use databases such as UniProt, OMIM, DisGeNET, and Open Targets to mine disease-target associations
- Understand the concept of target druggability and selectivity profiles

WEEK 2

Target Validation Techniques

- Distinguish between genetic, biochemical, and cell-based target validation approaches
- Study gene knockdown/knockout strategies
- Explore proteomics and transcriptomics tools for expression-level validation
- Understand biomarker identification and how validated targets translate into disease endpoints
- Introduction to structure determination techniques
- Retrieve and analyse 3D protein structures using the RCSB Protein Data Bank (PDB)

WEEK 3

Foundations of Computer-Aided Drug Design (CADD)

- Understand the two major CADD branches: Structure-Based Drug Design (SBDD) and Ligand-Based Drug Design (LBDD)
- Set up your computational environment
- Learn protein preparation workflows: adding hydrogens, assigning charges, and removing water molecules
- Introduction to small molecule representation: SMILES, InChI, and SDF formats
- Explore chemical databases
- Understand force fields and their role in molecular modelling

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 4

Lead Discovery - Molecular Docking & Virtual Screening

- Understand the principles of molecular recognition: shape complementarity, hydrogen bonding, and hydrophobic interactions
- Perform protein active site identification and binding pocket analysis
- Conduct grid-based molecular docking and Glide
- Screen a small compound library against your validated target and interpret docking scores
- Introduction to pharmacophore modelling using LigandScout or Phase
- Filter hits using Lipinski's Rule of Five and lead-like criteria

WEEK 5

Lead Optimisation - SAR and Medicinal Chemistry Principles

- Understand Structure-Activity Relationships (SAR) and how chemical modifications alter potency and selectivity
- Study bioisosteric replacement, scaffold hopping, and fragment growing strategies
- Learn R-group analysis and matched molecular pair (MMP) analysis
- Introduction to Free Energy Perturbation (FEP) concepts for relative binding affinity prediction
- Explore prodrug strategies and how they address bioavailability challenges
- Analyse case studies

WEEK 6

ADMET Prediction & Drug-Like Properties

- Understand ADMET: Absorption, Distribution, Metabolism, Excretion, and Toxicity in drug development
- Use SwissADME, pkCSM, and ADMETlab to predict pharmacokinetic profiles of lead compounds
- Study key parameters
- Introduction to hERG toxicity prediction and cardiotoxicity flagging
- Understand pan-assay interference compounds (PAINS) and how to identify and remove them

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 7

Molecular Dynamics & Advanced Simulation Techniques

- Understand the theory of Molecular Dynamics (MD) simulations and their role in drug design
- Set up and run short MD simulations
- Analyse MD trajectories
- Introduction to MM-GBSA and MM-PBSA methods for binding free energy estimation
- Explore induced-fit docking and allosteric drug design concepts
- Study ensemble docking to account for protein flexibility and conformational sampling

WEEK 8

Integrated Project & Industry Readiness

- Execute a full end-to-end mini drug design project
- Prepare a lead compound reports
- Introduction to intellectual property in drug discovery: patents, data exclusivity, and freedom to operate
- Understand regulatory pathways: IND filing, pre-clinical requirements, and early-phase clinical trial design
- Explore career tracks: medicinal chemist, computational chemist, pharmacologist, and drug discovery scientist

Adyapan School

WHO THIS IS FOR

This course is perfect for

- Students & Career Switchers
- Pharmacy, Biotechnology & Medicinal Chemistry Students
- Computational Chemistry Professionals
- Healthcare & Pharmaceutical Industry Professionals
- Bioinformatics & Molecular Modeling Enthusiasts
- Researchers & Lab Professionals

CERTIFICATIONS

ALUMNI NETWORK

Our alumni work at world-class companies

Amazon | Adobe | Google | Autodesk | Microsoft | Deloitte

Your career switch is one click away.

Ready to begin? Apply at adyapanschool.com or email us at support@adyapan.com

Apply Now